

# RONAK PATEL

☎ 8866044981 | ✉ [ronak.embedded84@gmail.com](mailto:ronak.embedded84@gmail.com) | 📍 Pune, India

## TECHNICAL SUMMARY

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- **Embedded Software Engineer with +10 years of work experience in Software and Hardware development.**
- Expertise in AUTOSAR architecture (MCAL, BSW, RTE integration and validation).
- Experience in communication protocol (CAN) and diagnostics protocol UDS (ISO 14229).
- Proficient in bootloaders (MCUboot, Vector) for secure firmware flashing and validation.
- Skilled in peripheral driver development and low-level debugging using Embedded C.
- Experience in ECU bring-up, hardware-software integration and debugging.
- Good understanding of embedded hardware design, including schematic capture and multi-layer PCB development.
- Experience in hardware development lifecycle (HDLC) and software development lifecycle (SDLC), Agile methodology and version control systems (Git).

## TOOLS

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- **EDA Tools:** Altium Designer, OrCAD, Eagle, CAM350
- **Programming Languages:** Embedded C, CAPL scripting
- **Automotive Grade ECU:** RH850, TC38X, TC39X, S32K148
- **AUTOSAR Tools:** Vector Davinci Configurator and Developer, EB Tresos Studio, ETAS RTA-CAR, CANoe, CANalyzer, PCAN, VFlash
- **Development IDEs:** AURIX Development Studio, S32 Design Studio IDE, e<sup>2</sup> studio
- **Compilers:** GHS, Tasking, GCC
- **Debugging Tools:** Lauterbach Trace32, Tasking iC7min, Logic Analyzer, Oscilloscope
- **Version Control Tool:** Git, Bitbucket
- **Project & Task Management:** JIRA

## WORK EXPERIENCE

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### RealThings India Private Limited – Pune, Maharashtra

*Sr. Software Engineer*

*July 2022 – Present*

- Supported MCAL, BSW, RTE integration using EB Tresos, Vector and ETAS stack toolchains.
- Worked on RTE configuration and ECU software integration based on AUTOSAR architecture
- Configured and Integrated COM stack, Memory stack (NvM, FEE, FLS), Diagnostic stack (DCM, DEM), System services (EcuM, BswM, ComM).
- Managed ECU Extract (ARXML/DBC) and Diagnostic Extract (CDD) and signal mapping for CAN communication.
- Implemented secure bootloader (Vector, MCUboot) with UDS-based firmware flashing and validation.
- Performed system-level debugging using CANoe, CANalyzer, and CAPL scripting
- Worked with embedded C for low-level driver development and AUTOSAR integration.

**Teq Diligent Product Solutions Pvt Ltd – Ahmedabad, Gujarat**  
*Embedded Engineer*

*Oct 2021 – July 2022*

- Performed schematic capture, PCB layout, and footprint creation using Altium Designer and OrCAD.
- Designed 6-layer high-speed PCBs with EMI/EMC compliance and thermal optimization.
- Managed complete hardware development lifecycle from schematic design to manufacturing release.
- Designed analog and digital circuits for embedded systems applications.
- Used oscilloscope and debugging tools for hardware–software validation and troubleshooting.
- Developed embedded firmware in Embedded C for hardware validation and peripheral interfacing.
- Coordinated with DFM, DFT teams, customers, and software engineers for design validation and issue resolution.

**Pandav Technology Pvt Ltd – Ahmedabad, Gujarat**  
*Engineering Solution Supervisor*

*Jan 2017 – Oct 2021*

- Designed schematics and multi-layer PCBs, including assembly support, board bring-up, debugging, and functional validation.
- Managed BOM optimization, vendor coordination, and production support activities.
- Supported testing and validation of biometric and embedded devices for production deployment.
- Delivered embedded ESP32 platforms with MQTT-based firmware and communication interfaces.

**Pandav Electrosoft – Ahmedabad, Gujarat**  
*Engineering Solution Engineer*

*April 2015 – Jan 2017*

- Supported development and technical activities in optoelectronics production processes.
- Performed repair, maintenance, and troubleshooting of biometric and embedded systems hardware.
- Assisted electronics production teams with PCB validation and field-level hardware support.

## **EDUCATION QUALIFICATION**

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**B.E. - Electronics & Communication Engineering**  
Gujarat Technological University, Ahmedabad, Gujarat, India

*April 2010 – July 2014*

## **PERSONAL PROFILE**

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| <b>Name:</b>            | Ronak Ravjibhai Patel    |
| <b>Date of Birth:</b>   | 08/04/1993               |
| <b>Sex:</b>             | Male                     |
| <b>Marital Status:</b>  | Married                  |
| <b>Nationality:</b>     | Indian                   |
| <b>Languages Known:</b> | Gujarati, Hindi, English |